2017 Harris Govern Training Conference

Texas Proud

2017 SQL INTERMEDIATE

Friday, November 10, 2017

HARRIS GOVERN

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This course will review more complex queries and how users can use these function to pull data from PACS tables. The session is structured for users who have existing SQL knowledge.

Agenda

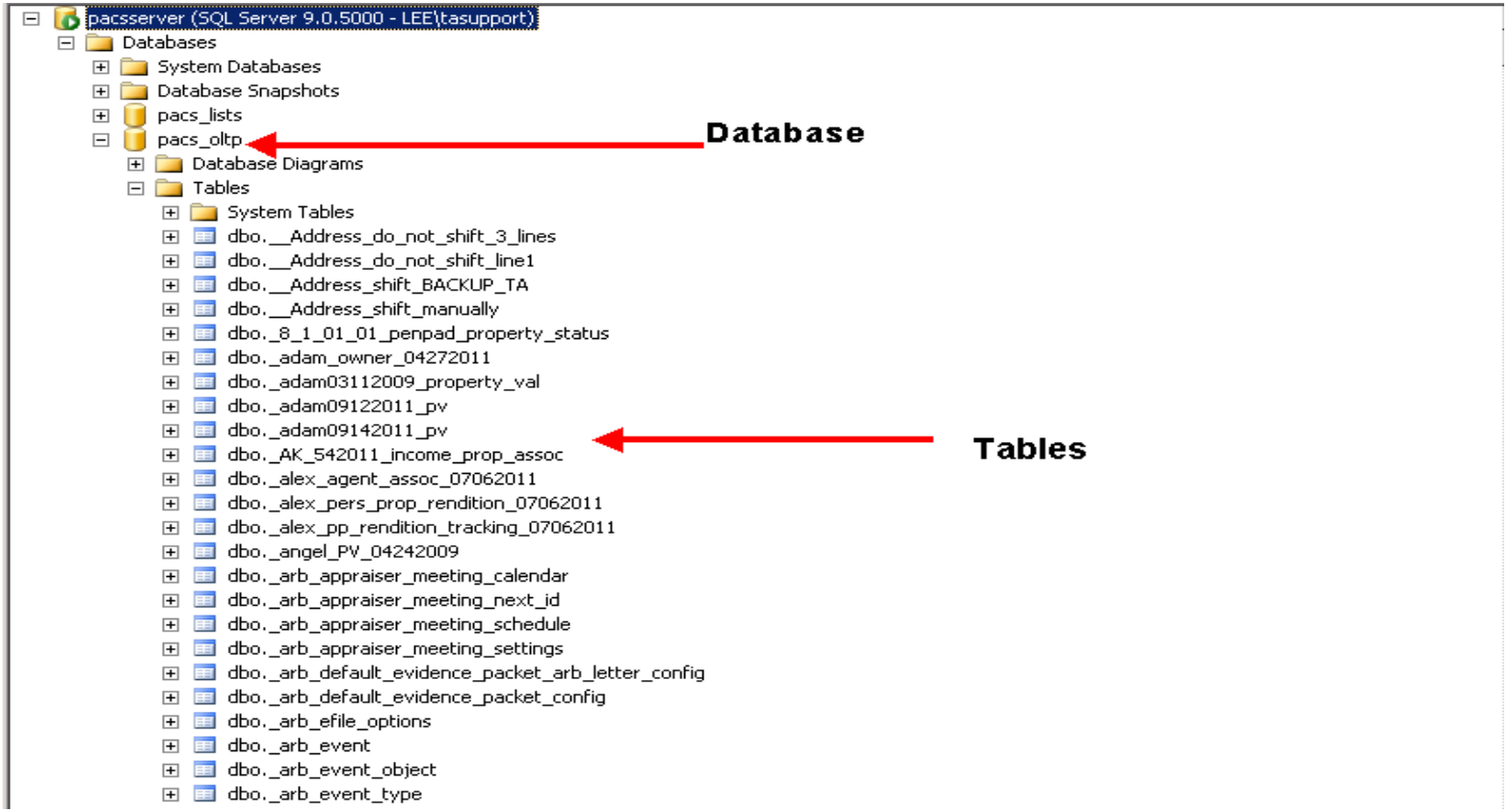
- Overview
- Joining Multiple tables
- Alias Table Names
- Using Sub queries
- Calculated fields
- Writing Query Demo

Overview

- Tables
- Columns and Data Types
- Rows
- Primary Keys

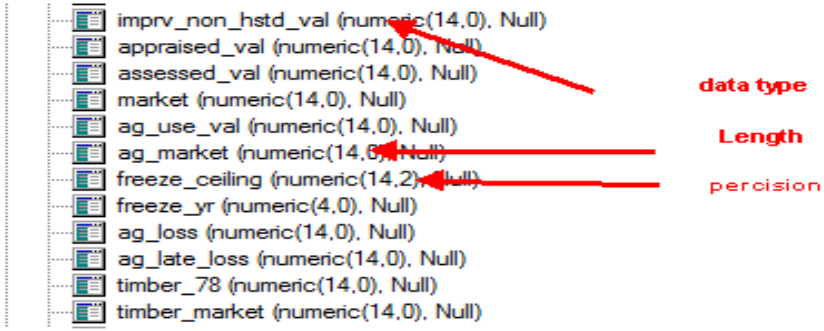
Tables

A table is a structure file that can store data of a specific type.



Columns and Data types

- A column will also contain a length. The length will tell the number of bytes a column can contain. There are certain types of Data types that do not require you to put a length, these are dynamic and they will grow in length as the data grows. Example of these would be int, datetime
- Other data types such as numeric will require you to give it a precision number as well, this is usually done when we are setting up a numeric field, and an example of this would be Land_non_hstd_val it will be a numeric data type, with a length of 14 and precision of 0, or you can see the acreage is setup as numeric with a length of 14 and precision of 4.
- Example:



Rows

Clipboard | Font | Alignment | Numb

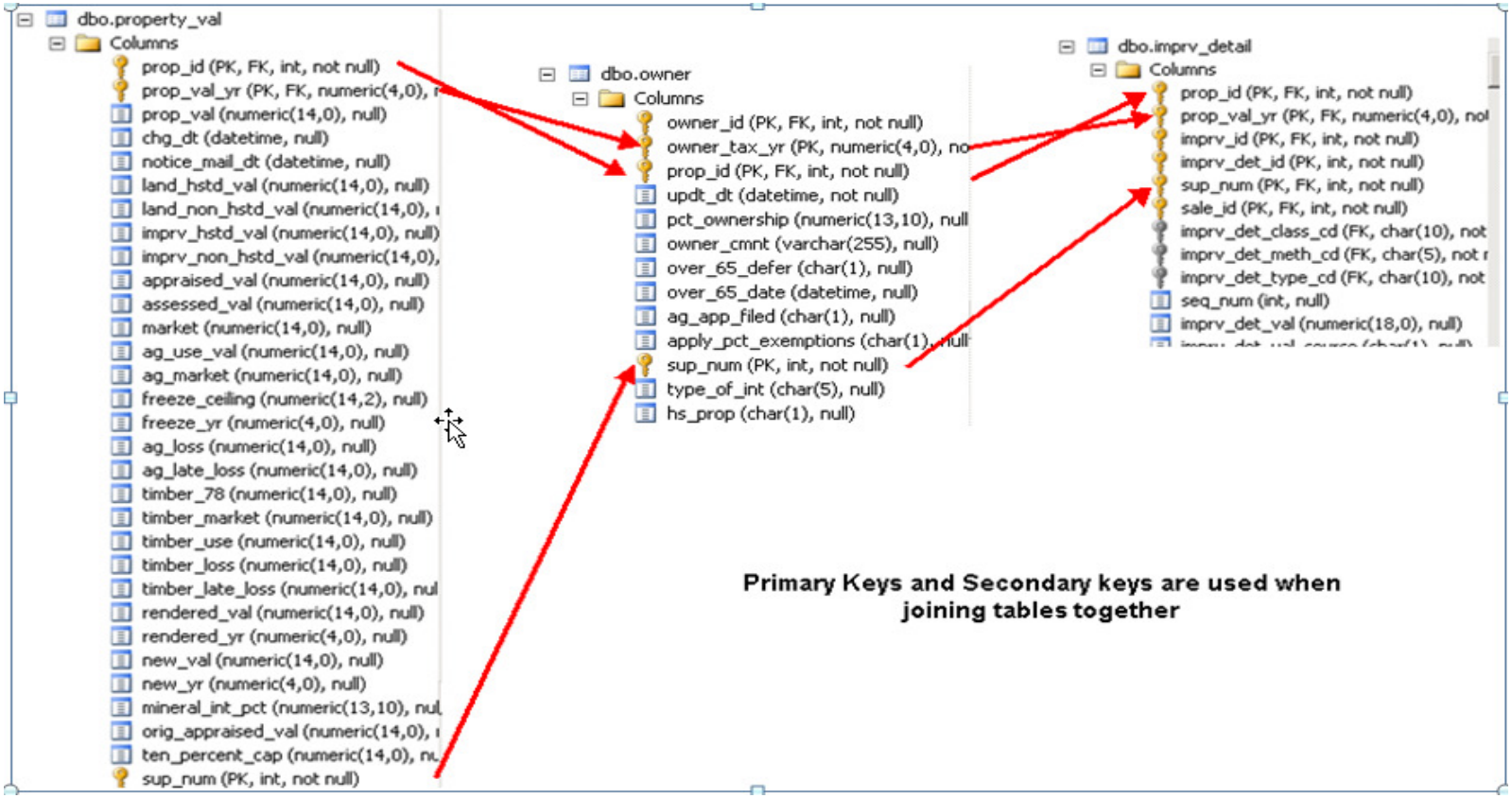
A1 | prop_id

	A	B	C	D	E	F	G
1	prop_id	imprv_id	imprv_adj_amt	imprv_adj_type_cd	land_seg_id	land_value	land_seg_adj_type
2	14160	1687	$ 1.00	DRP			
3	14298	743020	$ (78,878.00)	HIST			
4	14323	2220	$ (15,210.00)	DOR			
5	14337	743042	$ (301,314.00)	HIST			
6	14343				1928	$ (4,958.00)	HIST
7	14343	747791	$ (83,111.00)	HIST			
8	14345				94001	$ (72,000.00)	HIST
9	14345	743050	$ (1,124,422.00)	HIST			
10	14354	735102	$ (641,660.00)	HIST			
11	14412				1996	$ (50,000.00)	HIST
12	14412	2511	$ (189,288.00)	HIST			
13	14412	2511	$ (39,576.00)	HIST			
14	14413	2513	$ (73,059.00)	HIST			
15	14612	3232	$ (75,475.00)	HIST			
16	16085				3946	$ 181,209.00	CONV
17	17771				6011	$ (544,500.00)	DOR
18	17771	743258	$ (3,986,497.00)	DOR			
19	17828	3217	$ (27,999.00)	HIST			
20	17968	743297	$ (157,792.00)	HIST			
21	18073				6370	$ (20,909.00)	DOR
22	18073	72008	$ (90,119.00)	DOR			
23	18278	743396	$ (455,000.00)	MULT			
24	18346				6663	$ (60,480.00)	DOR

- Data in a table is stored in rows: each record saved is stored in its own row. The information stored in each row will contain unique information about that data.
- The columns are the fields in the table.
- The rows contain the data from PACS

Primary Keys

Primary Keys : A column or set of columns that uniquely identify all the rows in a table. Primary keys do not allow null values. No two rows can have the same primary key value; therefore a primary key value always uniquely identifies a single row. More than one key can uniquely identify rows in a table; each of these keys is called a **Secondary Key.**



Joining Tables

- Understanding Joins
- Inner Joins
- Outer Joins
- Joining Multiple tables
- Self Joins

Understanding Joins

- By using joins, you can retrieve data from two or more tables based on logical relationships between the tables
- A join condition defines the way two tables are related in a query by:
  - Specifying the column from each table to be used for the join. A typical join condition specifies a primary key from one table and its associated key in the other table.
  - Specifying a logical operator (for example, = or <>,) to be used in comparing values from the columns.

- Lets Break this down.

Inner Join and Outer Join

- **Inner Join** – An inner join will allow you to join multiple tables together to return the results that you are looking for. With an inner join it will only return the values in which the WHERE clause is all true
- **Outer Join** – An outer join will allow you to join a table and return a value in the field you asking to retrieve even if all the rows in all other tables are not related.
  - **Left outer join** - A left outer join is a join in which each element of the first collection is returned, regardless of whether it has any correlated elements in the second collection.

Alias Table Names

- An alias is an alternate name that can be used to make a connection.
  - We use alias names to make it easier to write SQL statements, when joining tables that contain the same column name.

Example:

Property_val pv

Owner o

Example of Joins using Alias

```
select
    a.file_as_name,addr.addr_line1,addr.addr_line2,addr.addr_line3,addr.addr_city,addr.a
    ddr_state,addr.addr_zip,
pv.legal_desc,s.situs_display
from property_val pv
inner join owner o
    on  pv.prop_id = o.prop_id
    and pv.prop_val_yr = o.owner_tax_yr
    and pv.sup_num = o.sup_num
inner join account a
    on o.owner_id = a.acct_id
inner join address addr
     on a.acct_id = addr.acct_id
left outer join situs s
    on pv.prop_id = s.prop_id
Where pv.prop_val_yr = 2012
And pv.prop_inactive_dt is null
order by a.file_as_name
```

DEMO

Using Sub Queries

- A subquery is an SQL SELECT statement that's nested inside of another SQL statement.
- The second form of subquery syntax, using In or Not In, allows you to use a set of values (rather than a single value) in a WHERE clause.

Example : Need to find all properties in one entity not in another entity.

```
Select prop_id from entity_prop_assoc
Where exmpt_tax_yr = 2012
And entity_id = 12
And prop_id not in (Select prop_id from entity_prop_assoc
Where exmpt_tax_yr = 2012
And entity_id = 14)
```

Calculated fields

- Creating a calculated fields allows you to use mathematical equations to return a value. For example, if we are wanting to add the imprv_hstd_val + imprv_non_hstd_val, Or we might need to get to base tax due in collections query for accounts that are delinquent. We can do this with in the query itself.

Example of a Calculated field

```
Select distinct prop_id, sum ((bill_adj_m_n_o + bill_adj_i_n_s) –
    (bill_m_n_o_pd + bill_i_n_s_pd + underage_mno_pd +
    underage_ins_pd + discount_mno_pd + discount_ins_pd)
        + (refund_m_n_o_pd + refund_i_n_s_pd)) as base_tax_due
From bill with (nolock)
Where active_bill = 'T' and coll_status_cd <> 'RS'
Group by prop_id
Having sum ((bill_adj_m_n_o + bill_adj_i_n_s) – (bill_m_n_o_pd +
    bill_i_n_s_pd + underage_mno_pd + underage_ins_pd +
    discount_mno_pd + discount_ins_pd)
        + (refund_m_n_o_pd + refund_i_n_s_pd)) > 0
```

Example of a Calculated field

```
select distinct pv.prop_id, p.geo_id, ac.file_as_name, pv.legal_desc,
    ld.state_cd, (pv.land_hstd_val + pv.land_non_hstd_val + pv.timber_market + pv.ag_market) as
    land_mkt,
    (pv.imprv_hstd_val + pv.imprv_non_hstd_val) as imprv_mkt
    from property_val pv with (nolock)
    inner join land_detail ld with (nolock) on
    pv.prop_id = ld.prop_id
    and pv.prop_val_yr = ld.prop_val_yr
    and pv.sup_num = ld.sup_num
    and ld.sale_id = 0
    inner join property p with (nolock) on
    p. Prop_id = pv.prop_id
    inner join owner o with (nolock) on
    o.prop_id = pv.prop_id
    and o.owner_tax_yr = pv.prop_val_yr
    and o.sup_num = pv.sup_num
    inner join account ac with (nolock) on
    ac.acct_id = o.owner_id
    where pv.prop_val_yr = 2012
    and (pv.prop_inactive_dt is null or udi_parent = 'T')
    order by p.geo_id
```

Writing Queries

Demo

Questions & Answers

Thank You For Attending This Session Please remember to fill out your surveys.